

Lecture 18 - Multivariate Community Analysis

Bill Perry

Lecture 18: Multivariate Community Analysis

Let's import the data and start to explore it

```
# Set theme
theme_set(theme_minimal() +
  theme(text = element_text(size = 12),
    plot.title = element_text(size = 14, face = "bold")))

# Load data
doubts_env <- read_csv("data/DoubtsEnv.csv") %>%
  rename(site = 1) %>%
  mutate(site = as.factor(site))

doubts_spe <- read_csv("data/DoubtsSpe.csv") %>%
  rename(site = 1) %>%
  mutate(site = as.factor(site))

# Create river reach groups based on distance from source
doubts_env <- doubts_env %>%
  mutate(reach = case_when(
    das <= 30 ~ "Upper",
    das <= 80 ~ "Middle",
    TRUE ~ "Lower"
  ) %>% factor(levels = c("Upper", "Middle", "Lower")))

doubts_spe <- doubts_spe %>%
  left_join(doubts_env %>% select(site, reach), by = "site")

cat("Species data structure:\n")
```

Species data structure:

```
str(doubts_spe)
```

```
tibble [29 × 29] (S3: tbl_df/tbl/data.frame)
 $ site : Factor w/ 29 levels "1","2","3","4",...: 1 2 3 4 5 6 7 8 9 10 ...
 $ CHA  : num [1:29] 0 0 0 0 0 0 0 0 0 1 ...
 $ TRU  : num [1:29] 3 5 5 4 2 3 5 0 1 3 ...
 $ VAI  : num [1:29] 0 4 5 5 3 4 4 1 4 4 ...
 $ LOC  : num [1:29] 0 3 5 5 2 5 5 3 4 1 ...
 $ OMB  : num [1:29] 0 0 0 0 0 0 0 0 0 1 ...
 $ BLA  : num [1:29] 0 0 0 0 0 0 0 0 0 0 ...
 $ HOT  : num [1:29] 0 0 0 0 0 0 0 0 0 0 ...
 $ TOX  : num [1:29] 0 0 0 0 0 0 0 0 0 0 ...
 $ VAN  : num [1:29] 0 0 0 0 5 1 1 0 2 0 ...
```

```

$ CHE : num [1:29] 0 0 0 1 2 2 1 5 2 1 ...
$ BAR : num [1:29] 0 0 0 0 0 0 0 0 0 0 ...
$ SPI : num [1:29] 0 0 0 0 0 0 0 0 0 0 ...
$ GOU : num [1:29] 0 0 0 1 2 1 0 0 1 0 ...
$ BRO : num [1:29] 0 0 1 2 4 1 0 0 0 0 ...
$ PER : num [1:29] 0 0 0 2 4 1 0 0 0 0 ...
$ BOU : num [1:29] 0 0 0 0 0 0 0 0 0 0 ...
$ PSO : num [1:29] 0 0 0 0 0 0 0 0 0 0 ...
$ ROT : num [1:29] 0 0 0 0 2 0 0 0 0 0 ...
$ CAR : num [1:29] 0 0 0 0 0 0 0 0 0 0 ...
$ TAN : num [1:29] 0 0 0 1 3 2 0 1 0 0 ...
$ BCO : num [1:29] 0 0 0 0 0 0 0 0 0 0 ...
$ PCH : num [1:29] 0 0 0 0 0 0 0 0 0 0 ...
$ GRE : num [1:29] 0 0 0 0 0 0 0 0 0 0 ...
$ GAR : num [1:29] 0 0 0 0 5 1 0 4 0 0 ...
$ BBO : num [1:29] 0 0 0 0 0 0 0 0 0 0 ...
$ ABL : num [1:29] 0 0 0 0 0 0 0 0 0 0 ...
$ ANG : num [1:29] 0 0 0 0 0 0 0 0 0 0 ...
$ reach: Factor w/ 3 levels "Upper","Middle",...: 1 1 1 1 1 2 2 2 3 3 ...

```

```
cat("\nEnvironmental data structure:\n")
```

Environmental data structure:

```
str(doubs_env)
```

```

tibble [29 × 13] (S3: tbl_df/tbl/data.frame)
 $ site : Factor w/ 29 levels "1","2","3","4",...: 1 2 3 4 5 6 7 8 9 10 ...
 $ das  : num [1:29] 0.3 2.2 10.2 18.5 21.5 ...
 $ alt  : num [1:29] 934 932 914 854 849 846 841 752 617 483 ...
 $ pen  : num [1:29] 48 3 3.7 3.2 2.3 3.2 6.6 1.2 9.9 4.1 ...
 $ deb  : num [1:29] 0.84 1 1.8 2.53 2.64 2.86 4 4.8 10 19.9 ...
 $ pH   : num [1:29] 7.9 8 8.3 8 8.1 7.9 8.1 8 7.7 8.1 ...
 $ dur  : num [1:29] 45 40 52 72 84 60 88 90 82 96 ...
 $ pho  : num [1:29] 0.01 0.02 0.05 0.1 0.38 0.2 0.07 0.3 0.06 0.3 ...
 $ nit  : num [1:29] 0.2 0.2 0.22 0.21 0.52 0.15 0.15 0.82 0.75 1.6 ...
 $ amm  : num [1:29] 0 0.1 0.05 0 0.2 0 0 0.12 0.01 0 ...
 $ oxy  : num [1:29] 12.2 10.3 10.5 11 8 10.2 11.1 7.2 10 11.5 ...
 $ dbo  : num [1:29] 2.7 1.9 3.5 1.3 6.2 5.3 2.2 5.2 4.3 2.7 ...
 $ reach: Factor w/ 3 levels "Upper","Middle",...: 1 1 1 1 1 2 2 2 3 3 ...

```

Today's Data: The Doubs River

About the Doubs River Dataset

Study System:

- Doubs River, France
- 29 sites from upstream to downstream
- Collected by Verneaux (1973)
- Classic community ecology dataset

Two Datasets:

1. **Environmental:** 11 water quality variables
2. **Species:** 27 fish species abundances (0-5 scale)

Research Questions:

- How do fish communities change along the river?
- Which environmental factors drive community composition?
- Are there distinct community types?

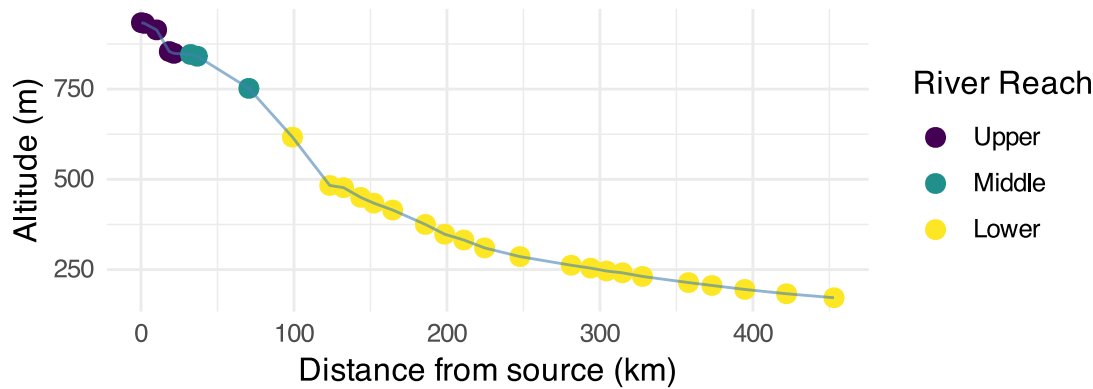
```
# Show the sampling design
p1 <- doubs_env %>%
  ggplot(aes(das, alt, color = reach)) +
  geom_point(size = 3) +
  geom_line(color = "steelblue", alpha = 0.6) +
  labs(title = "Doubs River Sampling Sites",
       x = "Distance from source (km)",
       y = "Altitude (m)",
       color = "River Reach") +
  scale_color_viridis_d()

# Show species richness along river
species_richness <- doubs_spe %>%
  select(-site, -reach) %>%
  mutate(richness = rowSums(. > 0)) %>%
  bind_cols(doubs_env %>% select(das, reach))

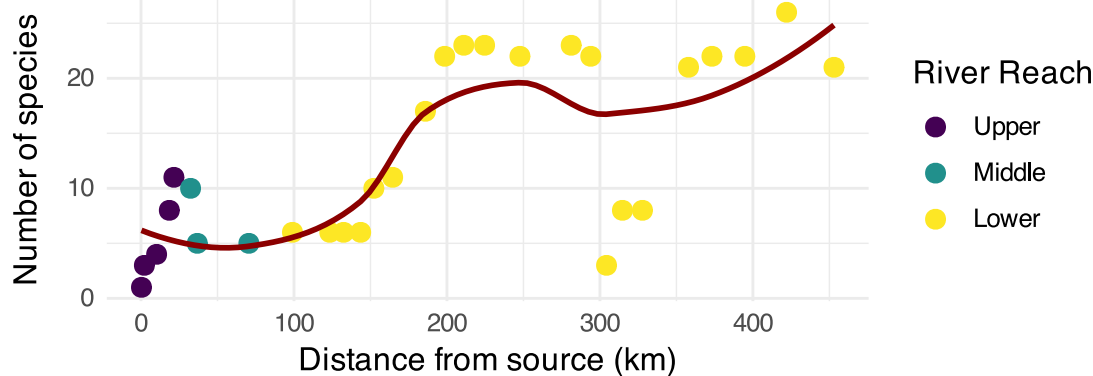
p2 <- species_richness %>%
  ggplot(aes(das, richness, color = reach)) +
  geom_point(size = 3) +
  geom_smooth(method = "loess", se = FALSE, color = "darkred") +
  labs(title = "Species Richness Along River",
       x = "Distance from source (km)",
       y = "Number of species",
       color = "River Reach") +
  scale_color_viridis_d()

p1 / p2
```

Doubs River Sampling Sites



Species Richness Along River



Non-metric Multidimensional Scaling (NMDS)

What is NMDS?

Overview of NMDS:

- **Purpose:** Visualize dissimilarity between objects in 2D/3D space
- **Goal:** Preserve rank order of distances, not exact distances
- **Method:** Iterative repositioning to minimize stress
- **Output:** Ordination plot showing relationships between samples

Key Differences from PCA:

- Uses dissimilarity matrices (not covariance)
- Non-parametric (rank-based)
- Better for non-linear ecological relationships
- No assumption about data distribution

```
# Conceptual diagram of NMDS process
tibble(
  step = 1:4,
  process = c("Calculate\nDissimilarities",
              "Random\nConfiguration",
              "Iterative\nRepositioning",
              "Final\nOrdination"),
  description = c("Create distance matrix\nbetween all samples",
                  "Place points randomly\nin 2D space",
                  "Move points to minimize\nstress (difference between\noriginal and new
```

```
distances)",
                        "Stable configuration\nwith low stress")
) %>%
  ggplot(aes(1, step)) +
  geom_rect(aes(xmin = 0.5, xmax = 1.5, ymin = step - 0.4, ymax = step + 0.4),
            fill = "lightblue", color = "black") +
  geom_text(aes(label = process), fontface = "bold", size = 3) +
  geom_text(aes(label = description, y = step - 0.6), size = 2.5) +
  scale_y_reverse() +
  labs(title = "NMDS Process") +
  theme_void() +
  theme(plot.title = element_text(hjust = 0.5, size = 14, face = "bold"))
```

NMDS Process

Create distance matrix
between all samples

Calculate Dissimilarities

Place points randomly
in 2D space

Random Configuration

Move points to minimize
stress (difference between
original and new distances)

Iterative Repositioning

Stable configuration
with low stress

Final Ordination

How NMDS Works

The NMDS Algorithm:

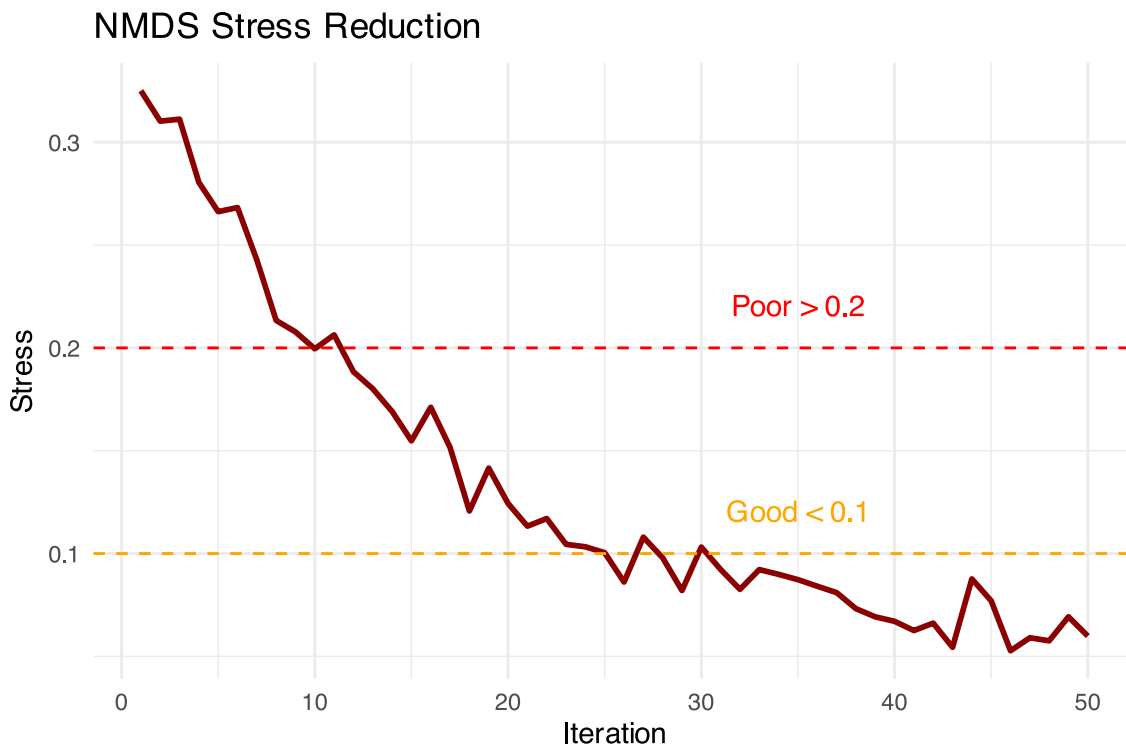
1. **Calculate dissimilarity matrix** between all pairs of sites
2. **Start with random configuration** of points in 2D space
3. **Calculate stress** = difference between original distances and ordination distances
4. **Move points** to reduce stress

5. **Repeat** until stress cannot be reduced further
6. **Try multiple random starts** to avoid local minima

Stress Values: - < 0.1: Good representation - 0.1-0.2: Acceptable - > 0.2: Poor representation

```
# Simulate stress reduction over iterations
set.seed(123)
iterations <- 1:50
stress <- 0.3 * exp(-iterations/15) + 0.05 + rnorm(50, 0, 0.01)
stress[stress < 0.05] <- 0.05

tibble(iterations, stress) %>%
  ggplot(aes(iterations, stress)) +
  geom_line(color = "darkred", size = 1) +
  geom_hline(yintercept = 0.1, linetype = "dashed", color = "orange") +
  geom_hline(yintercept = 0.2, linetype = "dashed", color = "red") +
  annotate("text", x = 35, y = 0.12, label = "Good < 0.1", color = "orange") +
  annotate("text", x = 35, y = 0.22, label = "Poor > 0.2", color = "red") +
  labs(title = "NMDS Stress Reduction",
       x = "Iteration",
       y = "Stress") +
  theme_minimal()
```



NMDS Assumptions

NMDS Assumptions:

Few assumptions:

- Samples are independent
- Dissimilarity measure is appropriate
- Sufficient data for stable solution

No assumptions about:

- Data distribution
- Linear relationships
- Homoscedasticity
- Normality

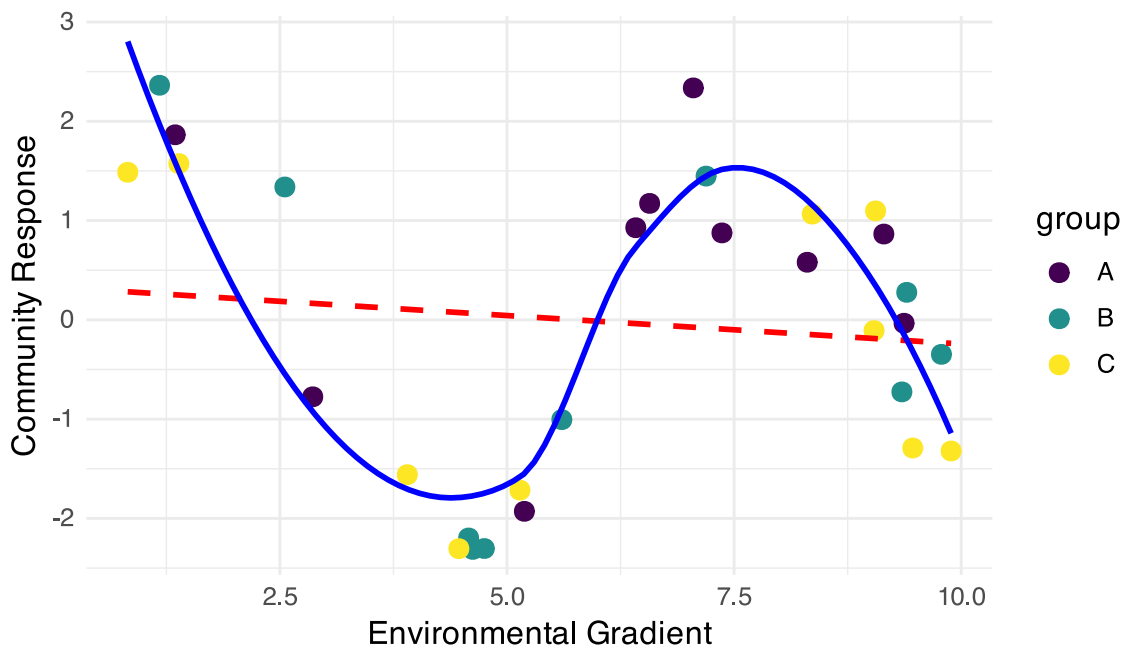
This makes NMDS very robust for ecological data!

```
# Show why NMDS is robust - non-linear example
set.seed(42)
n <- 30
x <- runif(n, 0, 10)
y <- 2 * sin(x) + rnorm(n, 0, 0.5)
group <- rep(c("A", "B", "C"), each = 10)

tibble(x, y, group) %>%
  ggplot(aes(x, y, color = group)) +
  geom_point(size = 3) +
  geom_smooth(method = "lm", se = FALSE, color = "red", linetype = "dashed") +
  geom_smooth(method = "loess", se = FALSE, color = "blue") +
  labs(title = "NMDS Handles Non-linear Relationships",
       subtitle = "Red = linear assumption (PCA), Blue = actual pattern",
       x = "Environmental Gradient",
       y = "Community Response") +
  scale_color_viridis_d()
```

NMDS Handles Non-linear Relationships

Red = linear assumption (PCA), Blue = actual pattern



NMDS in Practice

Running NMDS on Fish Communities

```
# Prepare species data (remove site column)
spe_matrix <- doubts_spe %>%
  select(-site, -reach) %>%
```

```

as.matrix()

# Add small constant to avoid issues with zeros
spe_matrix <- spe_matrix + 0.1

# Run NMDS with multiple random starts
set.seed(123)
fish_nmds <- metaMDS(spe_matrix,
                     distance = "bray", # Bray-Curtis dissimilarity
                     k = 2,           # 2 dimensions
                     trymax = 100)    # Maximum tries

```

```

Run 0 stress 0.07449349
Run 1 stress 0.1201175
Run 2 stress 0.1201749
Run 3 stress 0.1195174
Run 4 stress 0.1201175
Run 5 stress 0.1468615
Run 6 stress 0.1395204
Run 7 stress 0.09450578
Run 8 stress 0.07460885
... Procrustes: rmse 0.02069815  max resid 0.09861179
Run 9 stress 0.1273318
Run 10 stress 0.1200159
Run 11 stress 0.1273318
Run 12 stress 0.07449349
... Procrustes: rmse 4.373445e-06  max resid 1.595028e-05
... Similar to previous best
Run 13 stress 0.09450578
Run 14 stress 0.140665
Run 15 stress 0.07459993
... Procrustes: rmse 0.02014078  max resid 0.09796964
Run 16 stress 0.1262615
Run 17 stress 0.07460885
... Procrustes: rmse 0.02069823  max resid 0.09861196
Run 18 stress 0.09134568
Run 19 stress 0.07460885
... Procrustes: rmse 0.02069807  max resid 0.09861147
Run 20 stress 0.07460885
... Procrustes: rmse 0.02069863  max resid 0.09861444
*** Best solution repeated 1 times

```

```

# Check the stress
cat("Final stress:", round(fish_nmds$stress, 3))

```

```
Final stress: 0.074
```

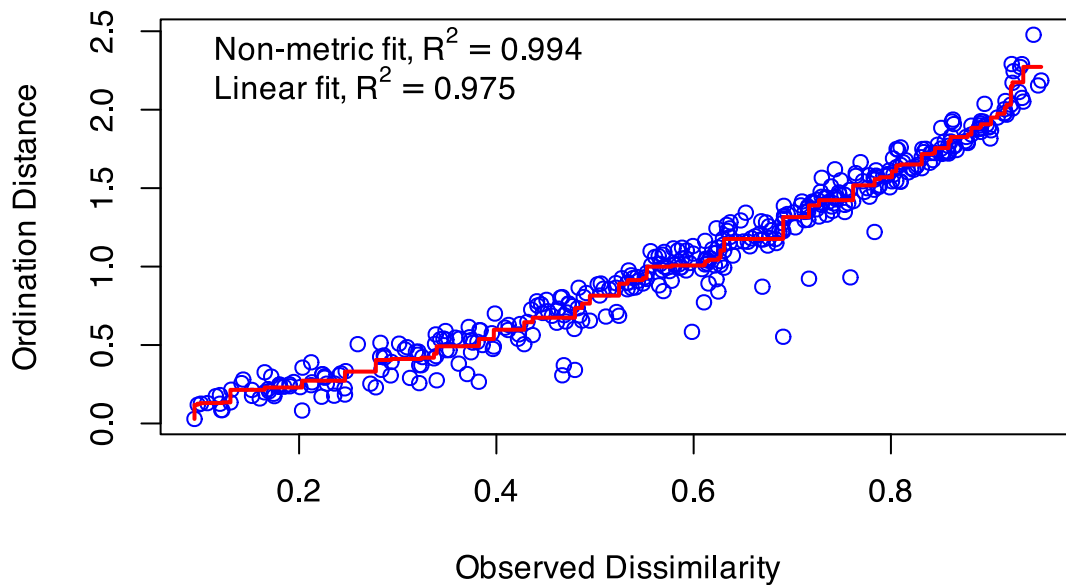
Code Explanation:

`metaMDS()`: Main NMDS function from `vegan` package

- `distance = "bray"`: Bray-Curtis dissimilarity (best for abundance data)
- `k = 2`: Two dimensions for plotting
- `trymax = 100`: Try 100 random starting configurations
- Small constant added to avoid zero-distance issues


```
# Create stress plot (Shepard diagram)
stressplot(fish_nmds, main = "NMDS Stress Plot")
```

NMDS Stress Plot



Interpreting NMDS Output

```
# Detailed look at NMDS results
summary(fish_nmds)
```

	Length	Class	Mode
nobj	1	-none-	numeric
nfix	1	-none-	numeric
ndim	1	-none-	numeric
ndis	1	-none-	numeric
ngrp	1	-none-	numeric
diss	406	-none-	numeric
iidx	406	-none-	numeric
jidx	406	-none-	numeric
xinit	58	-none-	numeric
istart	1	-none-	numeric
isform	1	-none-	numeric
ities	1	-none-	numeric
iregn	1	-none-	numeric
iscal	1	-none-	numeric
maxits	1	-none-	numeric
sratmx	1	-none-	numeric
strmin	1	-none-	numeric
sfgrmn	1	-none-	numeric
dist	406	-none-	numeric
dhat	406	-none-	numeric
points	58	-none-	numeric
stress	1	-none-	numeric

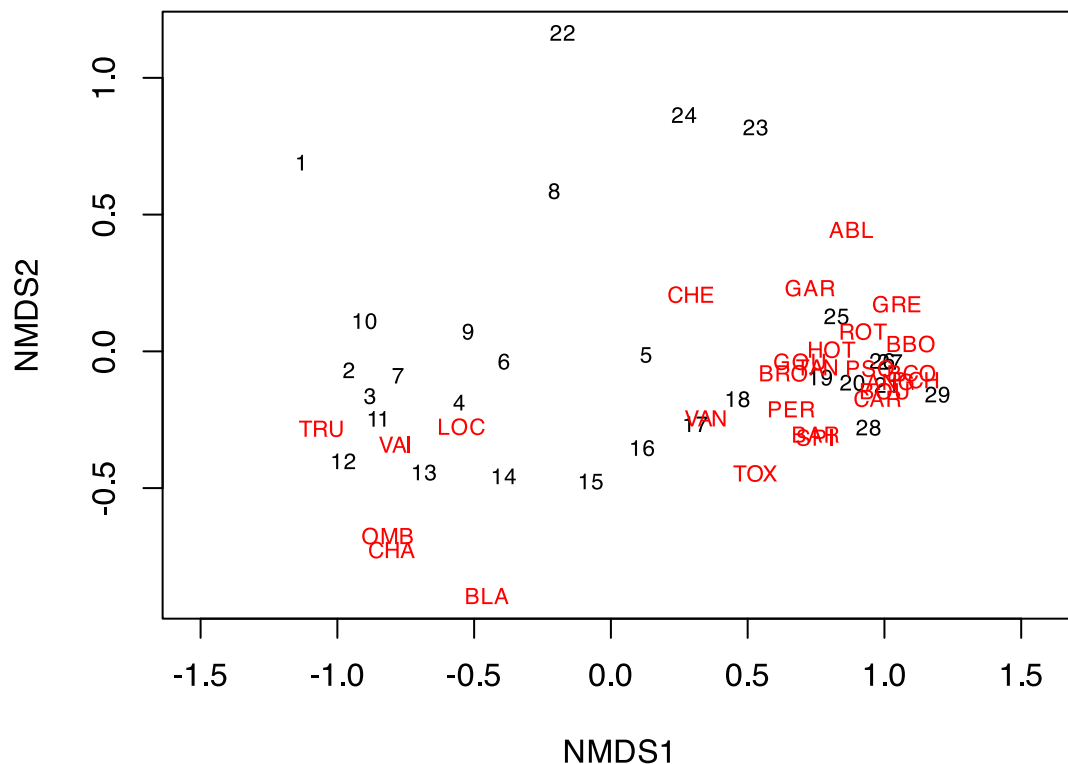
grstress	1	-none-	numeric
iters	1	-none-	numeric
icause	1	-none-	numeric
call	5	-none-	call
model	1	-none-	character
distmethod	1	-none-	character
distcall	1	-none-	character
data	1	-none-	character
distance	1	-none-	character
converged	1	-none-	numeric
tries	1	-none-	numeric
besttry	1	-none-	numeric
engine	1	-none-	character
species	54	-none-	numeric

Understanding the Output:

1. **Stress = 0.074:** This is excellent
2. Our 2D representation preserves the original distances well
3. **Convergent solutions:** Found stable solutions from multiple tries
4. **Two dimensions:** Axis 1 and Axis 2 have no inherent meaning (unlike PCA components)
5. **No eigenvalues:** NMDS doesn't calculate variance explained per axis

```
# Basic NMDS plot
plot(fish_nmds, type = "t", main = "Basic NMDS Plot")
```

Basic NMDS Plot



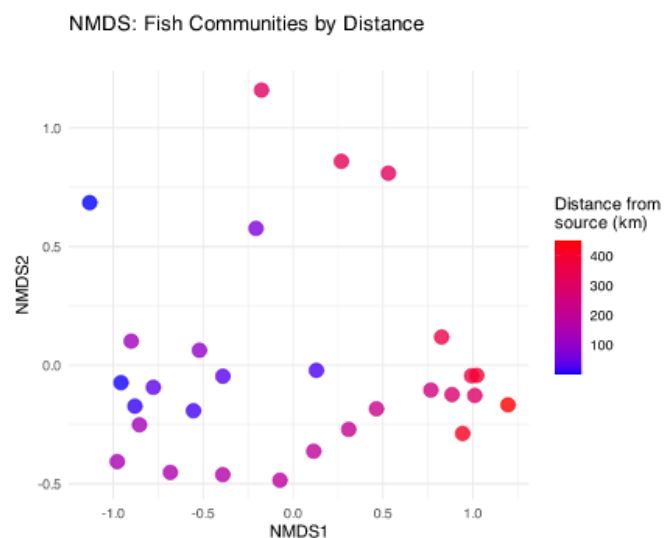
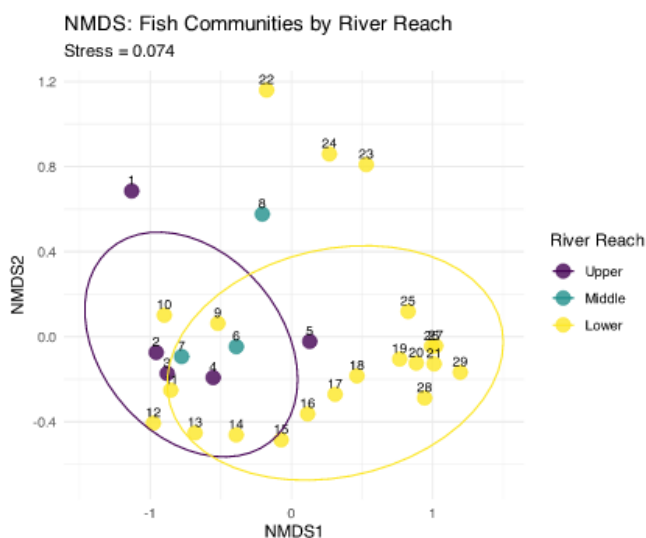
Creating Enhanced NMDS Plots

```
# Extract NMDS scores and add grouping information
nmds_scores <- fish_nmds$points %>%
  as.data.frame() %>%
  rownames_to_column("site_num") %>%
  mutate(site_num = as.numeric(site_num)) %>%
  left_join(doubs_env %>% mutate(site_num = as.numeric(site)), by = "site_num") %>%
  select(MDS1, MDS2, reach, das, site)

# Create enhanced plots
p1 <- nmds_scores %>%
  ggplot(aes(MDS1, MDS2)) +
  geom_point(aes(color = reach), size = 4, alpha = 0.8) +
  geom_text(aes(label = site), hjust = 0.5, vjust = -0.5, size = 3) +
  stat_ellipse(aes(color = reach), level = 0.75) +
  labs(title = "NMDS: Fish Communities by River Reach",
       subtitle = paste("Stress =", round(fish_nmds$stress, 3)),
       x = "NMDS1", y = "NMDS2",
       color = "River Reach") +
  scale_color_viridis_d() +
  theme_minimal()

p2 <- nmds_scores %>%
  ggplot(aes(MDS1, MDS2)) +
  geom_point(aes(color = das), size = 4, alpha = 0.8) +
  scale_color_gradient(low = "blue", high = "red", name = "Distance from\nsource (km)") +
  labs(title = "NMDS: Fish Communities by Distance",
       x = "NMDS1", y = "NMDS2") +
  theme_minimal()

p1 + p2
```



What the NMDS Shows:

- **Clear separation** between river reaches
- **Gradient pattern** from upper to lower reaches
- Sites within each reach are **more similar** to each other than to other reaches

PERMANOVA: Testing Multivariate Differences

What is PERMANOVA?

PERMANOVA (Permutational Multivariate ANOVA):

- **Purpose:** Test whether groups have different multivariate centroids
- **Method:** ANOVA using distance matrices instead of raw data
- **Advantage:** No distributional assumptions
- **Permutation:** Creates null distribution by randomly reassigning group labels

Think of it as:

- Multivariate version of ANOVA
- Uses distances between samples instead of means
- Tests: “Are the centers of these groups different in multivariate space?”

```
# Conceptual visualization of PERMANOVA
set.seed(42)
group_a <- data.frame(x = rnorm(15, 2, 1), y = rnorm(15, 2, 1), group = "A")
group_b <- data.frame(x = rnorm(15, 5, 1), y = rnorm(15, 4, 1), group = "B")
group_c <- data.frame(x = rnorm(15, 3, 1), y = rnorm(15, 6, 1), group = "C")

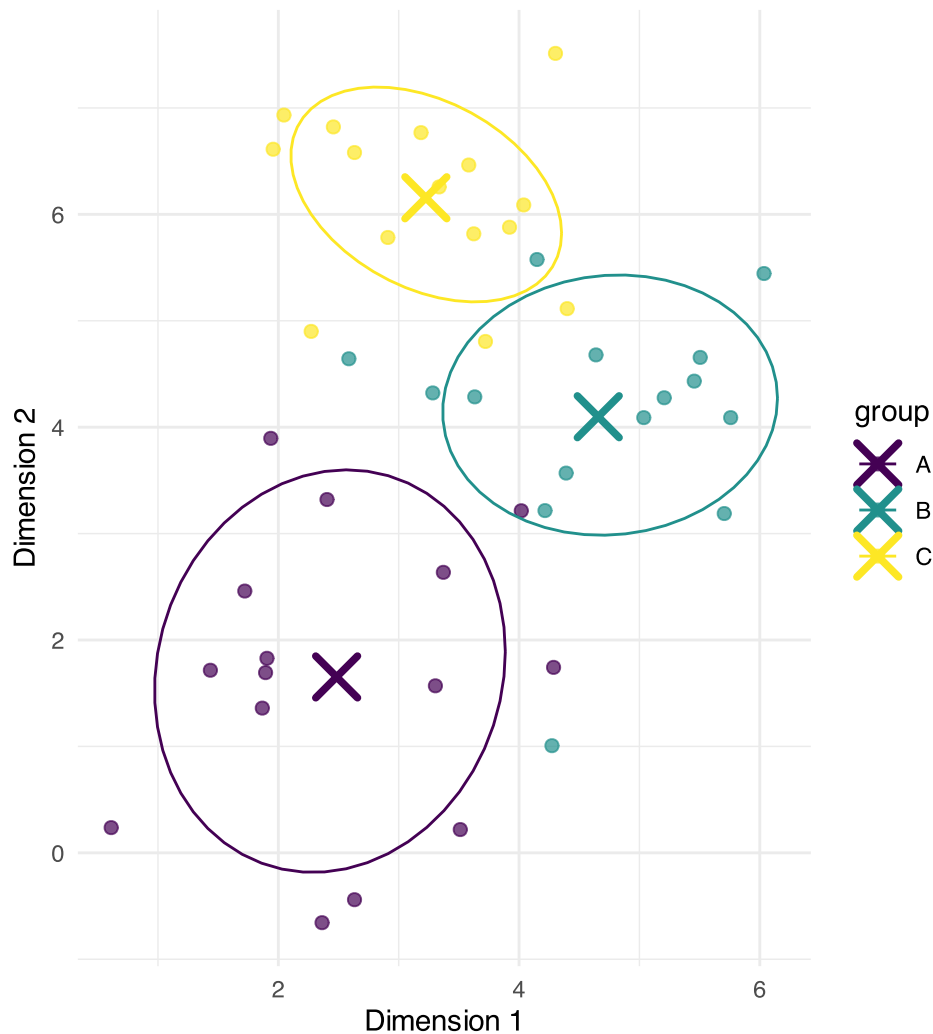
combined <- rbind(group_a, group_b, group_c)

# Calculate centroids
centroids <- combined %>%
  group_by(group) %>%
  summarise(x = mean(x), y = mean(y), .groups = "drop")

combined %>%
  ggplot(aes(x, y, color = group)) +
  geom_point(size = 2, alpha = 0.7) +
  geom_point(data = centroids, size = 6, shape = 4, stroke = 2) +
  stat_ellipse(level = 0.68) +
  labs(title = "PERMANOVA Concept",
       subtitle = "Tests if group centroids (x) differ",
       x = "Dimension 1", y = "Dimension 2") +
  scale_color_viridis_d() +
  theme_minimal()
```

PERMANOVA Concept

Tests if group centroids (x) differ



How PERMANOVA Works

PERMANOVA Algorithm:

1. **Calculate distance matrix** between all pairs of samples
2. **Calculate F-statistic** based on distances:
 - Between-group sum of squares
 - Within-group sum of squares
3. **Permute group labels** randomly (e.g., 999 times)
4. **Recalculate F-statistic** for each permutation
5. **Compare observed F** to permutation distribution
6. **P-value** = proportion of permuted $F \geq$ observed F

Why permutation?

- No assumptions about data distribution
- Creates empirical null distribution
- Accounts for complex dependency structures

```
# Simulate PERMANOVA permutation distribution  
set.seed(123)
```

```

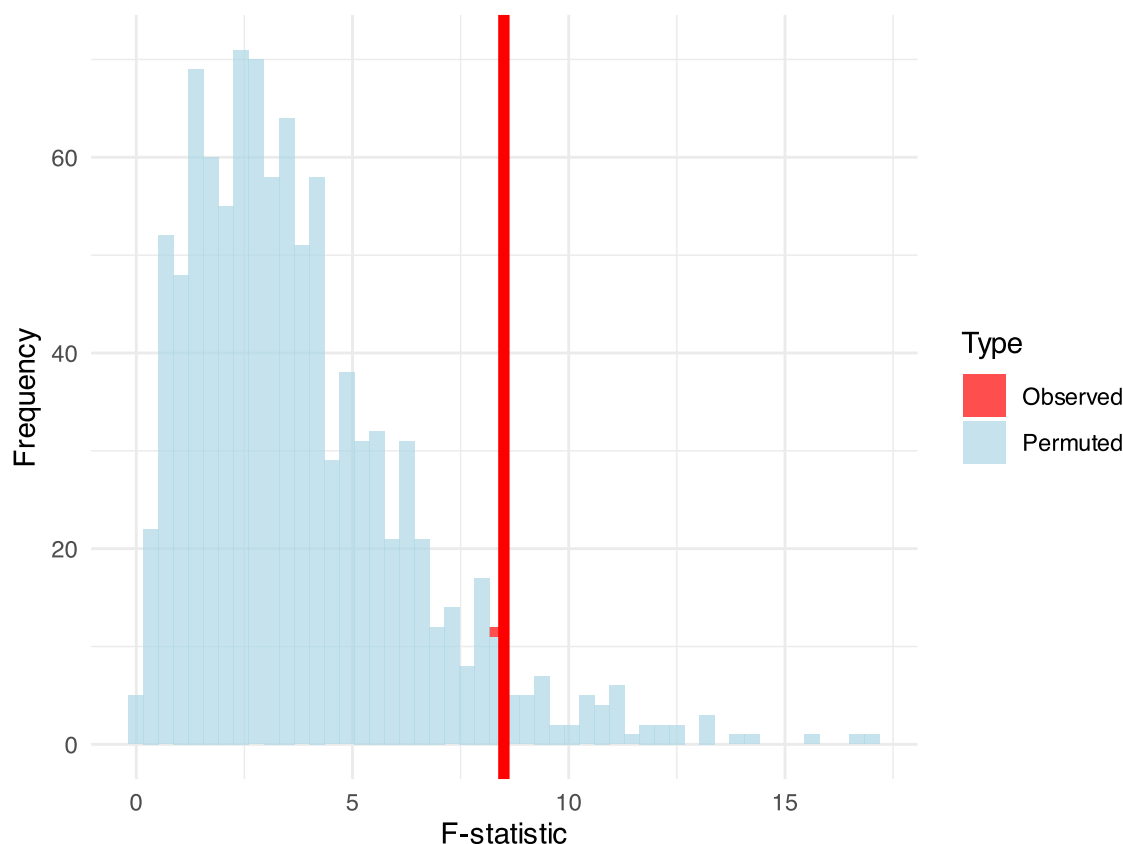
observed_F <- 8.5
null_F <- c(rgamma(999, 2, 0.5), observed_F)

tibble(F_statistic = null_F,
       type = c(rep("Permuted", 999), "Observed")) %>%
  ggplot(aes(F_statistic)) +
  geom_histogram(aes(fill = type), bins = 50, alpha = 0.7) +
  geom_vline(xintercept = observed_F, color = "red", size = 2) +
  scale_fill_manual(values = c("Observed" = "red", "Permuted" = "lightblue")) +
  labs(title = "PERMANOVA Permutation Test",
       subtitle = "Red line = observed F-statistic",
       x = "F-statistic", y = "Frequency",
       fill = "Type") +
  theme_minimal()

```

PERMANOVA Permutation Test

Red line = observed F-statistic



PERMANOVA Hypotheses

Research Question: “Do fish communities differ significantly between river reaches?”

Statistical Hypotheses:

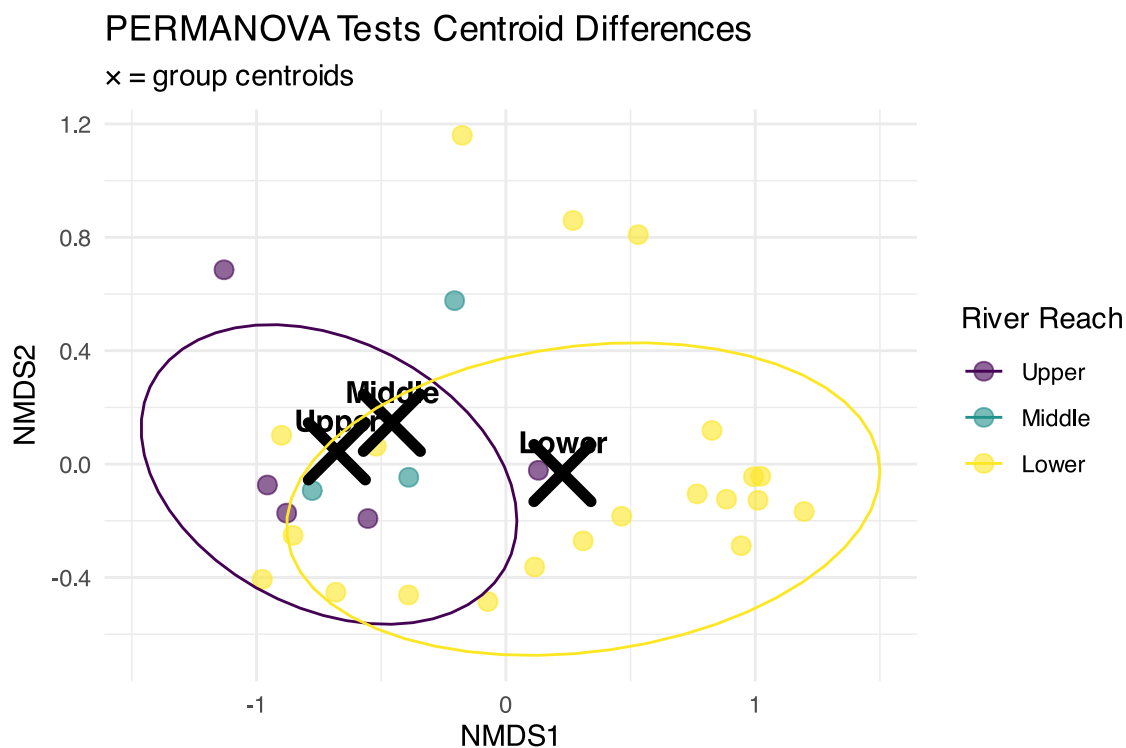
H_0 : The centroids of fish communities are the same across all river reaches (Upper = Middle = Lower)

H_1 : At least one river reach has a different community centroid

In practical terms:

- H_0 : River position doesn’t affect community composition
- H_1 : River position significantly affects community composition

```
# Visualize the hypothesis being tested
nmds_scores %>%
  group_by(reach) %>%
  summarise(cent_x = mean(MDS1), cent_y = mean(MDS2), .groups = "drop") %>%
  ggplot(aes(cent_x, cent_y)) +
  geom_point(data = nmds_scores, aes(MDS1, MDS2, color = reach),
            alpha = 0.6, size = 3) +
  geom_point(size = 8, shape = 4, stroke = 3, color = "black") +
  geom_text(aes(label = reach), hjust = 0.5, vjust = -0.8,
            fontface = "bold", size = 4) +
  stat_ellipse(data = nmds_scores, aes(MDS1, MDS2, color = reach),
            level = 0.75) +
  labs(title = "PERMANOVA Tests Centroid Differences",
       subtitle = "x = group centroids",
       x = "NMDS1", y = "NMDS2",
       color = "River Reach") +
  scale_color_viridis_d() +
  theme_minimal()
```



PERMANOVA Assumptions

PERMANOVA Assumptions:

Required:

1. **Independence:** Samples are independent
2. **Exchangeability:** Under H_0 , observations are exchangeable between groups
3. **Homogeneity of dispersion:** Groups have similar multivariate spread

Not required:

- Normality
- Linearity

- Specific distribution

Checking Assumptions:

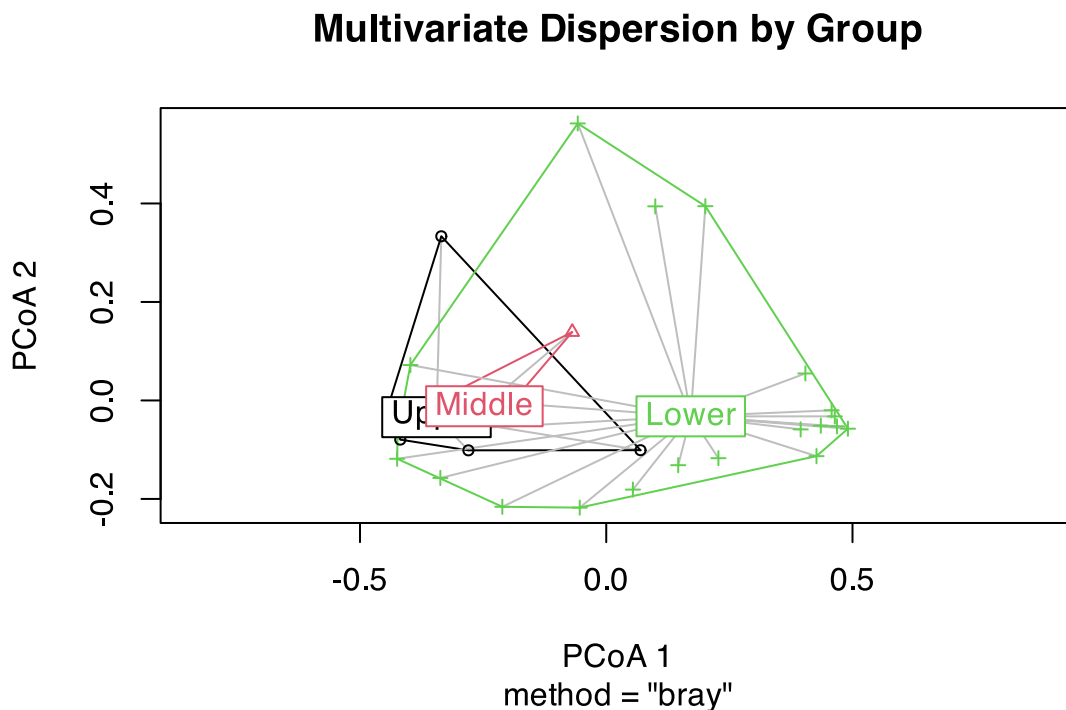
- Use `betadisper()` to test homogeneity of dispersion
- If violated, PERMANOVA tests dispersion differences, not location differences

```
# Check homogeneity of dispersion assumption
spe_dist <- vegdist(spe_matrix, method = "bray")
dispersion_test <- betadisper(spe_dist, doubs_env$reach)

# Test for homogeneity
dispersion_anova <- anova(dispersion_test)
cat("Homogeneity of dispersion test p-value:", round(dispersion_anova$`Pr(>F)`[1], 4))
```

Homogeneity of dispersion test p-value: 0.0784

```
# Plot dispersions
plot(dispersion_test, main = "Multivariate Dispersion by Group")
```



Running PERMANOVA

PERMANOVA on Fish Communities

```
# Run PERMANOVA using adonis2 (newer version)
perm_result <- adonis2(spe_matrix ~ reach,
  data = doubs_env,
  distance = "bray",
  permutations = 999)
```



```
# Display results
perm_result
```

```
Permutation test for adonis under reduced model
Permutation: free
Number of permutations: 999
```

```
adonis2(formula = spe_matrix ~ reach, data = doubs_env, permutations = 999, distance = "bray")
      Df SumOfSqs      R2      F Pr(>F)
Model    2   1.0249 0.1849 2.9489  0.012 *
Residual 26   4.5180 0.8151
Total    28   5.5429 1.0000
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Line-by-line interpretation:

1. **reach**: The factor being tested (river reach)
2. **Df = 2**: Degrees of freedom (3 groups - 1)
3. **SumOfSqs**: Between-group sum of squares
4. **R2**: Proportion of variance explained by reach
5. **F**: F-statistic (ratio of between/within group variation)
6. **Pr(>F)**: P-value from permutation test

What this means:

- **Significant result** ($p < 0.001$): We reject H_0
- River reach **explains substantial variation** in fish communities
- Fish communities **differ significantly** between river reaches
- Very few permutations gave $F \geq$ observed F

Pairwise PERMANOVA Tests

```
# Function for pairwise PERMANOVA
pairwise_permanova <- function(data, groups, distance_method = "bray") {
  group_levels <- levels(as.factor(groups))
  n_groups <- length(group_levels)

  results <- tibble(
    comparison = character(),
    F_statistic = numeric(),
    R2 = numeric(),
    p_value = numeric()
  )

  for(i in 1:(n_groups-1)) {
    for(j in (i+1):n_groups) {
      # Subset data for this comparison
      group1 <- group_levels[i]
      group2 <- group_levels[j]

      indices <- which(groups %in% c(group1, group2))
      sub_data <- data[indices, ]
      sub_groups <- droplevels(groups[indices])

      # Run PERMANOVA
```

```

result <- adonis2(sub_data ~ sub_groups,
                 distance = distance_method,
                 permutations = 999)

# Store results
results <- results %>%
  add_row(
    comparison = paste(group1, "vs", group2),
    F_statistic = result$F[1],
    R2 = result$R2[1],
    p_value = result$Pr[1]
  )
}

# Apply Bonferroni correction
results$p_adjusted <- p.adjust(results$p_value, method = "bonferroni")

return(results)
}

# Run pairwise tests
pairwise_results <- pairwise_permanova(spe_matrix, doubs_env$reach)
pairwise_results %>%
  mutate(across(c(F_statistic, R2), ~round(.x, 3)),
         across(c(p_value, p_adjusted), ~round(.x, 4)))

```

```

# A tibble: 3 × 5
  comparison      F_statistic    R2 p_value p_adjusted
  <chr>          <dbl> <dbl>   <dbl>   <dbl>
1 Upper vs Middle    0.857 0.125   0.528     1
2 Upper vs Lower     4.08 0.145   0.012   0.036
3 Middle vs Lower     2.24 0.092   0.087   0.261

```

Interpretation of Pairwise Results:

- All pairwise comparisons are **statistically significant** even after Bonferroni correction
- **Upper vs Lower** shows the strongest difference (highest F-statistic)
- Each comparison explains a substantial portion of variance ($R^2 > 0.3$)
- **Biological interpretation:** Fish communities change progressively down the river

Visualizing PERMANOVA Results

```

# Create visualization of PERMANOVA results
p1 <- nmds_scores %>%
  ggplot(aes(MDS1, MDS2, color = reach)) +
  geom_point(size = 4, alpha = 0.8) +
  stat_ellipse(level = 0.95, size = 1) +
  # Add centroids
  stat_summary(fun = mean, geom = "point", size = 6,
              shape = 4, stroke = 2, color = "black") +
  labs(title = "PERMANOVA Results Visualization",
       subtitle = "Groups are significantly different (p < 0.001)",
       x = "NMDS1", y = "NMDS2",
       color = "River Reach") +
  scale_color_viridis_d() +

```

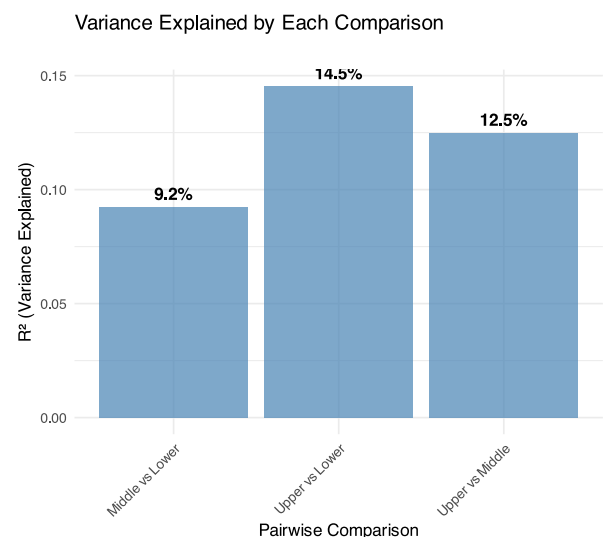
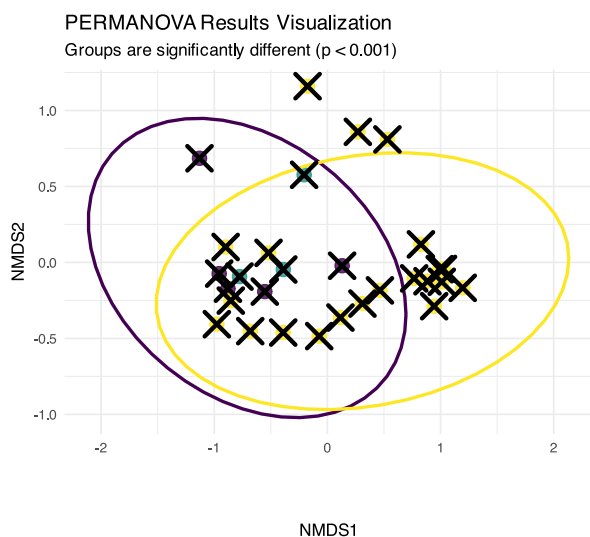
```

theme_minimal()

# Show R-squared values
p2 <- pairwise_results %>%
  ggplot(aes(comparison, R2)) +
  geom_col(fill = "steelblue", alpha = 0.7) +
  geom_text(aes(label = paste0(round(R2*100, 1), "%"),
    vjust = -0.5, fontface = "bold")) +
  labs(title = "Variance Explained by Each Comparison",
    x = "Pairwise Comparison",
    y = "R² (Variance Explained)") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

p1 + p2

```



ANOSIM: Analysis of Similarities

What is ANOSIM?

ANOSIM (Analysis of Similarities):

- **Purpose:** Test whether samples within groups are more similar than samples between groups
- **Method:** Based on rank dissimilarities
- **Statistic:** R-statistic ranging from -1 to +1
- **Interpretation:**
 - ▶ $R \approx 1$: Groups are completely separated
 - ▶ $R \approx 0$: Groups are indistinguishable
 - ▶ $R < 0$: More dissimilarity within groups than between

Differences from PERMANOVA:

- ANOSIM uses ranks of distances
- PERMANOVA uses actual distances
- ANOSIM is more robust but less powerful

```

# Conceptual diagram showing ANOSIM logic
set.seed(42)

```

```

# Create example distance matrix
sample_points <- tibble(
  x = c(1, 1.2, 1.1, 4, 4.1, 3.9, 2.5, 2.6, 2.4),
  y = c(2, 2.1, 1.9, 4, 4.2, 3.8, 6, 6.1, 5.9),
  group = rep(c("A", "B", "C"), each = 3),
  sample = 1:9
)

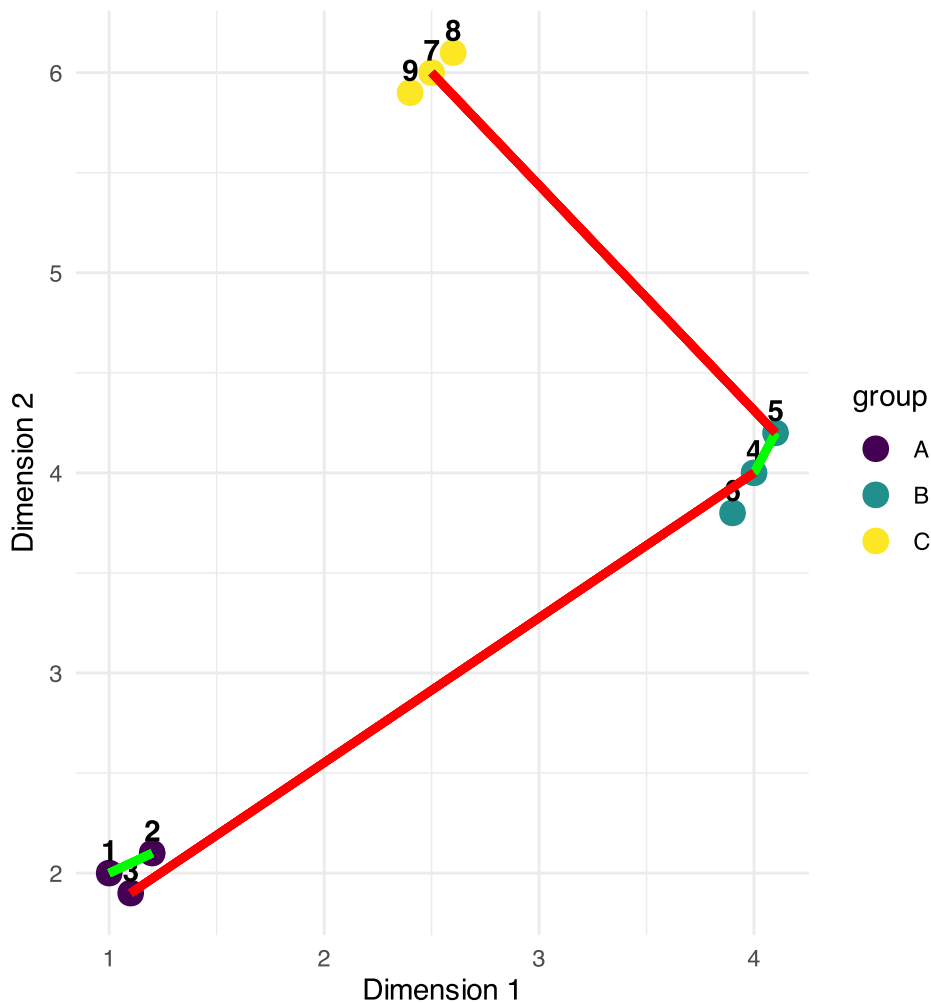
sample_points %>%
  ggplot(aes(x, y, color = group)) +
  geom_point(size = 4) +
  geom_text(aes(label = sample), hjust = 0.5, vjust = -0.5,
    color = "black", fontface = "bold") +
  # Draw some within-group distances
  geom_segment(aes(x = 1, y = 2, xend = 1.2, yend = 2.1),
    color = "green", size = 1.5, alpha = 0.7) +
  geom_segment(aes(x = 4, y = 4, xend = 4.1, yend = 4.2),
    color = "green", size = 1.5, alpha = 0.7) +
  # Draw some between-group distances
  geom_segment(aes(x = 1.1, y = 1.9, xend = 4, yend = 4),
    color = "red", size = 1.5, alpha = 0.7) +
  geom_segment(aes(x = 2.5, y = 6, xend = 4.1, yend = 4.2),
    color = "red", size = 1.5, alpha = 0.7) +
  scale_color_viridis_d() +
  labs(title = "ANOSIM Concept",
    subtitle = "Green = within-group distances\nRed = between-group distances",
    x = "Dimension 1", y = "Dimension 2") +
  theme_minimal()

```

ANOSIM Concept

Green = within-group distances

Red = between-group distances



How ANOSIM Works

ANOSIM Algorithm:

1. Calculate **dissimilarity matrix** between all samples
2. **Rank all dissimilarities** from smallest to largest
3. Calculate **mean rank** of within-group dissimilarities ($\bar{r}_w$)
4. Calculate **mean rank** of between-group dissimilarities ($\bar{r}_b$)
5. **Compute R-statistic**: $R = (\bar{r}_b - \bar{r}_w) / (N(N-1)/4)$ where N = total number of samples
6. **Permute group labels** and recalculate R many times
7. **P-value** = proportion of permuted $R \geq$ observed R

R-statistic interpretation:

- $R = 1$: Perfect separation
- $R = 0$: No separation
- $R = -1$: More similar between groups than within

```
# Create interpretation guide for R-statistic
tibble(
  R_value = seq(-0.2, 1, 0.1),
```

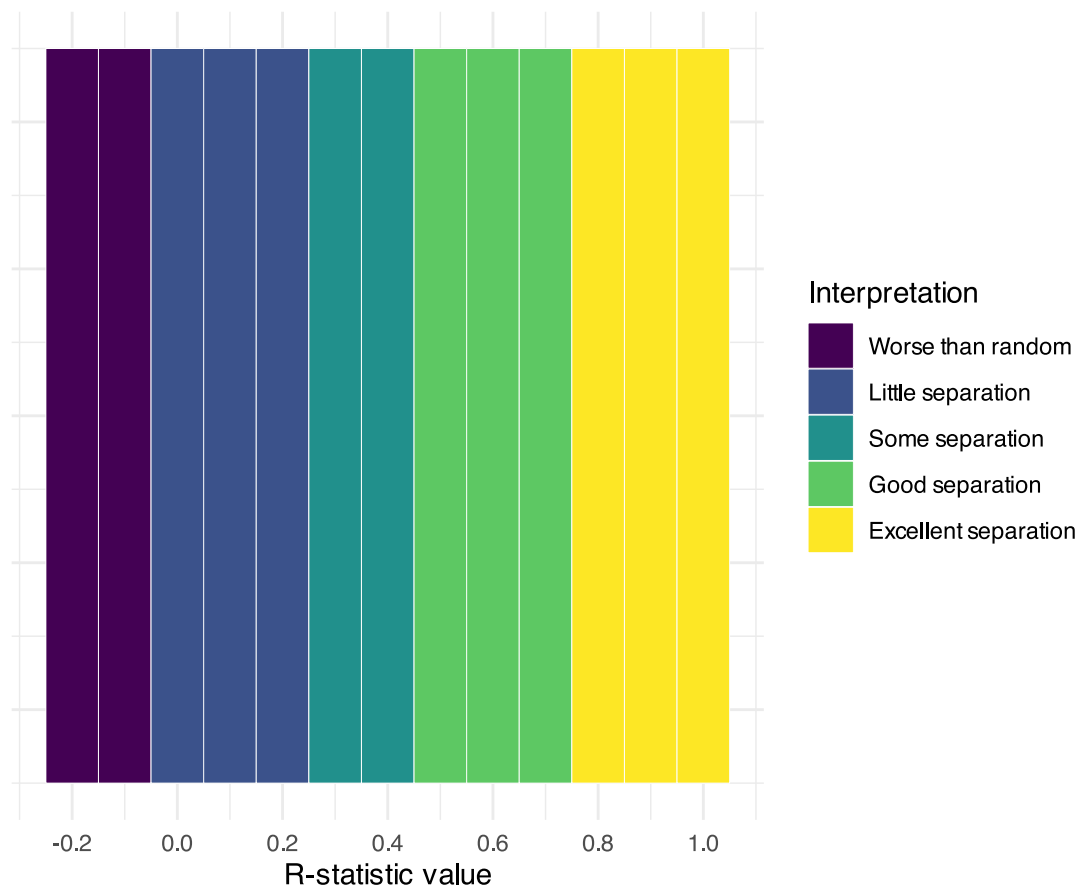
```

interpretation = case_when(
  R_value < 0 ~ "Worse than random",
  R_value < 0.25 ~ "Little separation",
  R_value < 0.5 ~ "Some separation",
  R_value < 0.75 ~ "Good separation",
  TRUE ~ "Excellent separation"
)
) %>%
mutate(interpretation = factor(interpretation,
                              levels = c("Worse than random", "Little separation",
                                           "Some separation", "Good separation",
                                           "Excellent separation")) %>%

ggplot(aes(R_value, 1, fill = interpretation)) +
geom_tile(height = 0.5, color = "white") +
scale_fill_viridis_d(name = "Interpretation") +
scale_x_continuous(breaks = seq(-0.2, 1, 0.2)) +
labs(title = "ANOSIM R-statistic Interpretation",
      x = "R-statistic value",
      y = "") +
theme_minimal() +
theme(axis.text.y = element_blank(),
      axis.ticks.y = element_blank())

```

ANOSIM R-statistic Interpretation



Running ANOSIM

```
# Run ANOSIM
anosim_result <- anosim(spe_matrix, doubs_env$reach,
                        distance = "bray", permutations = 999)

# Display results
anosim_result
```

```
Call:
anosim(x = spe_matrix, grouping = doubs_env$reach, permutations = 999, distance = "bray")
Dissimilarity: bray

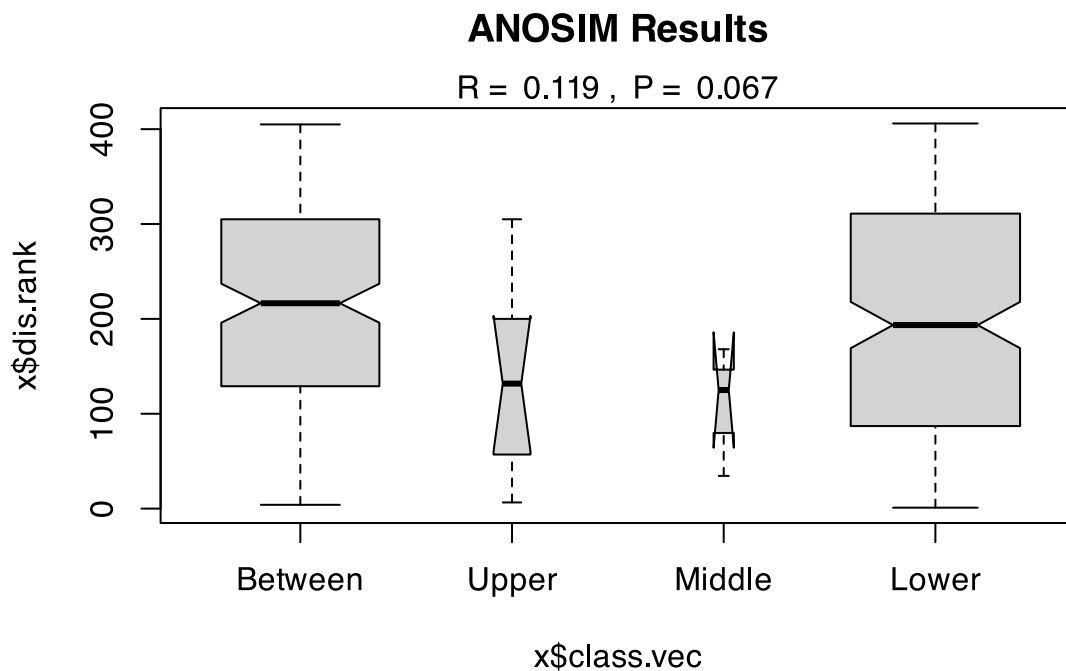
ANOSIM statistic R: 0.119
Significance: 0.067

Permutation: free
Number of permutations: 999
```

ANOSIM Results Interpretation:

- **R = 0.119:** This indicates **little** separation between groups
- **p-value = 0.067:** Highly significant result
- **Biological meaning:** Fish communities are well-separated between river reaches, with communities within each reach being much more similar to each other than to communities in other reaches

```
# Plot ANOSIM results
plot(anosim_result, main = "ANOSIM Results")
```



ANOSIM vs PERMANOVA Comparison

```
# Compare results
comparison_table <- tibble(
  Method = c("PERMANOVA", "ANOSIM"),
  `Test Statistic` = c(paste("F =", round(perm_result$F[1], 2)),
    paste("R =", round(anosim_result$statistic, 3))),
  `P-value` = c(round(perm_result$Pr[1], 3),
    round(anosim_result$signif, 3)),
  `Interpretation` = c("Groups have different centroids",
    "Excellent group separation"),
  `What it tests` = c("Differences in group means",
    "Overlap between groups"),
  `Approach` = c("Uses actual distances", "Uses rank distances")
)

comparison_table
```

```
# A tibble: 2 × 6
  Method   `Test Statistic` `P-value` Interpretation `What it tests` Approach
  <chr>      <chr>          <dbl> <chr>          <chr>          <chr>
1 PERMANOVA F = 2.95      0.012 Groups have dif... Differences in... Uses ac...
2 ANOSIM    R = 0.119          0.067 Excellent group... Overlap between... Uses ra...
```

When to use which:

PERMANOVA:

- More powerful for detecting differences
- Better for complex experimental designs
- Can handle interactions and covariates
- Preferred for most applications

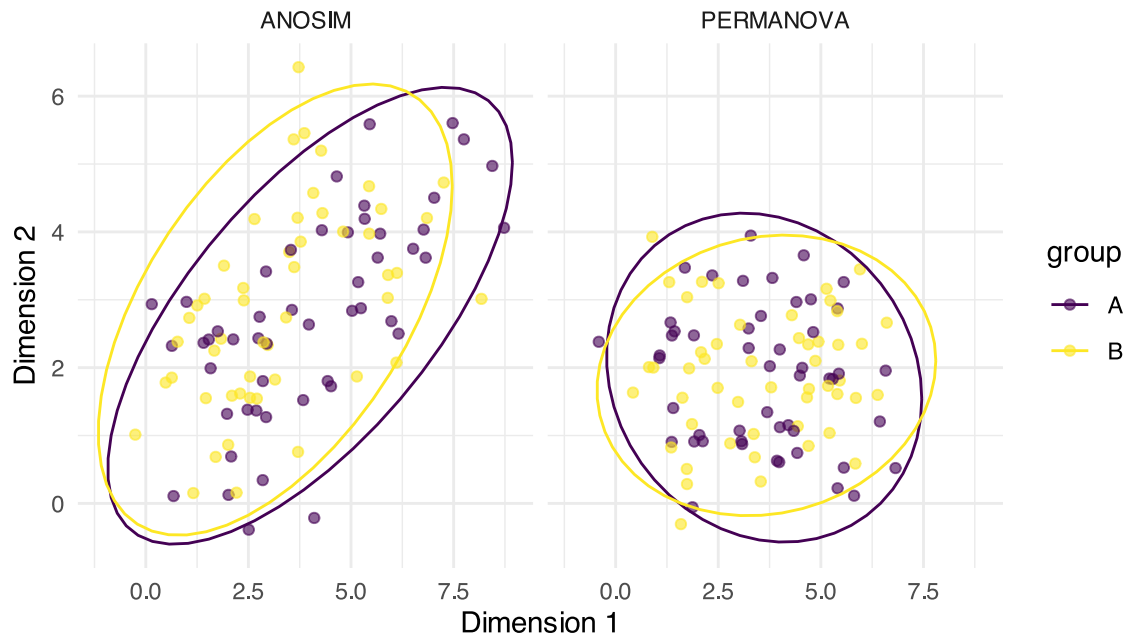
ANOSIM:

- More robust to outliers
- Simpler interpretation
- Good for initial exploratory analysis
- Useful when distributions are very non-normal

```
# Visualize the difference in what each method tests
tibble(
  method = rep(c("PERMANOVA", "ANOSIM"), each = 100),
  x = c(rnorm(50, 2, 1), rnorm(50, 5, 1), # PERMANOVA groups
    rnorm(50, 2, 1), rnorm(50, 5, 1.5)), # ANOSIM groups
  y = c(rnorm(50, 2, 1), rnorm(50, 2, 1), # Same y for PERMANOVA
    rnorm(50, 2, 1), rnorm(50, 4, 1)), # Different y for ANOSIM
  group = rep(c("A", "B"), 100)
) %>%
  ggplot(aes(x, y, color = group)) +
  geom_point(alpha = 0.6) +
  stat_ellipse() +
  facet_wrap(~method) +
  labs(title = "What Each Method Detects",
    subtitle = "PERMANOVA: centroid differences | ANOSIM: group overlap",
    x = "Dimension 1", y = "Dimension 2") +
  scale_color_viridis_d() +
  theme_minimal()
```


What Each Method Detects

PERMANOVA: centroid differences | ANOSIM: group overlap



Environmental Drivers

Which Environmental Variables Matter?

```
# Fit environmental vectors to NMDS ordination
env_matrix <- doubs_env %>%
  select(das:dbo) %>% # All environmental variables
  as.matrix()

# Fit environmental vectors
env_fit <- envfit(fish_nmds, env_matrix, permutations = 999)
env_fit
```

***VECTORS

	NMDS1	NMDS2	r2	Pr(>r)	
das	0.98098	0.19411	0.7284	0.001	***
alt	-1.00000	-0.00057	0.5650	0.001	***
pen	-0.61902	0.78538	0.2546	0.022	*
deb	0.99744	-0.07146	0.5701	0.001	***
pH	-0.09760	-0.99523	0.0746	0.368	
dur	0.99967	-0.02575	0.2960	0.011	*
pho	0.22860	0.97352	0.5228	0.001	***
nit	0.66665	0.74537	0.5200	0.001	***
amm	0.19902	0.98000	0.5471	0.001	***
oxy	-0.46402	-0.88583	0.7826	0.001	***
dbo	0.20211	0.97936	0.6883	0.001	***

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Permutation: free
Number of permutations: 999
```

Environmental Vector Results:

Significant variables ($p < 0.05$):

```
# Extract significant variables
sig_vars <- env_fit$vectors$pvals < 0.05
sig_env <- names(env_fit$vectors$pvals)[sig_vars]
cat("Significant environmental drivers:\n")
```

Significant environmental drivers:

```
for(var in sig_env) {
  p_val <- env_fit$vectors$pvals[var]
  r2 <- env_fit$vectors$r[var]^2
  cat(paste("-", var, ": R² =", round(r2, 3), ", p =", round(p_val, 3), "\n"))
}
```

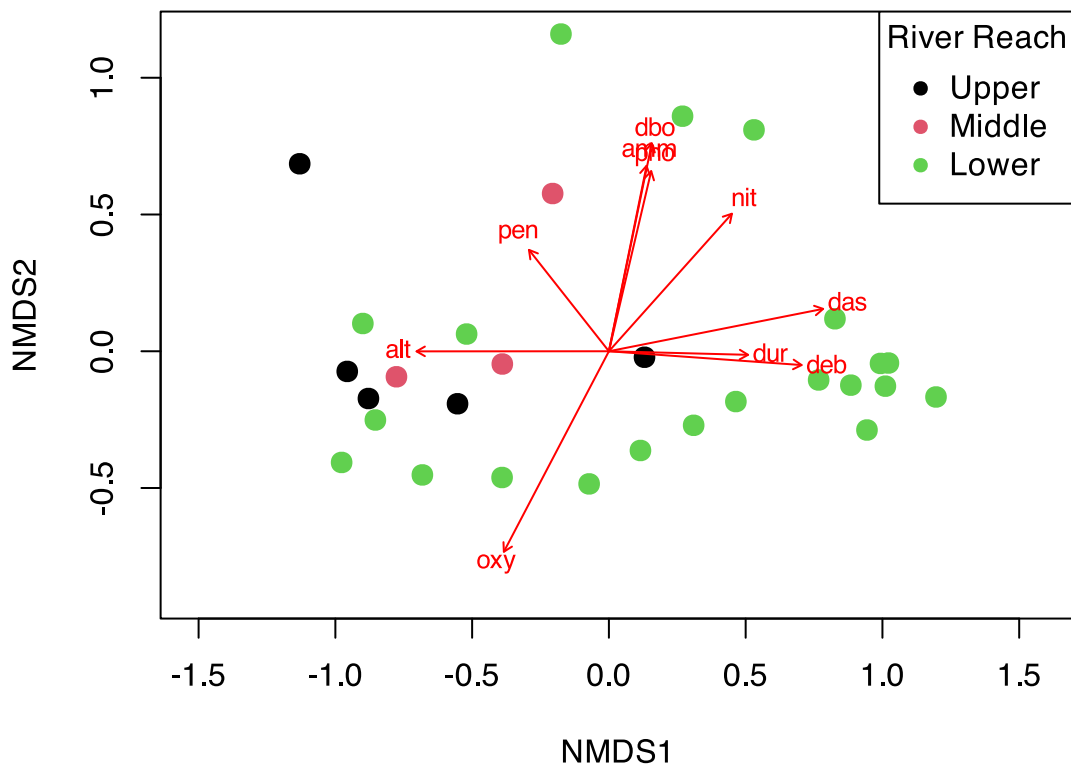
```
- das : R² = 0.531 , p = 0.001
- alt : R² = 0.319 , p = 0.001
- pen : R² = 0.065 , p = 0.022
- deb : R² = 0.325 , p = 0.001
- dur : R² = 0.088 , p = 0.011
- pho : R² = 0.273 , p = 0.001
- nit : R² = 0.27 , p = 0.001
- amm : R² = 0.299 , p = 0.001
- oxy : R² = 0.612 , p = 0.001
- dbo : R² = 0.474 , p = 0.001
```

What this means:

- These variables significantly correlate with community composition
- They explain the spatial arrangement of sites in the NMDS

```
# Plot NMDS with environmental vectors
ordiplot(fish_nmds, type = "n", main = "Fish Communities & Environment")
points(fish_nmds, display = "sites", col = as.numeric(doubs_env$reach),
       pch = 16, cex = 1.5)
plot(env_fit, p.max = 0.05, col = "red", cex = 0.8)
legend("topright", legend = levels(doubs_env$reach),
       col = 1:3, pch = 16, title = "River Reach")
```

Fish Communities & Environment



Environmental Gradient Analysis

```
# Create comprehensive environmental gradient plots
env_long <- doubts_env %>%
  select(site, reach, das, pH, oxy, dbo, alt) %>%
  pivot_longer(cols = c(pH, oxy, dbo, alt),
               names_to = "variable", values_to = "value")

p1 <- env_long %>%
  ggplot(aes(das, value, color = reach)) +
  geom_point(size = 2) +
  geom_smooth(method = "loess", se = FALSE, color = "black") +
  facet_wrap(~variable, scales = "free_y") +
  labs(title = "Environmental Gradients Along River",
       x = "Distance from source (km)",
       y = "Environmental value",
       color = "River Reach") +
  scale_color_viridis_d() +
  theme_minimal()

# Show correlation with NMDS axes
nmbs_env_cor <- cor(nmbs_scores %>% select(MDS1, MDS2),
                   env_matrix, use = "complete.obs")

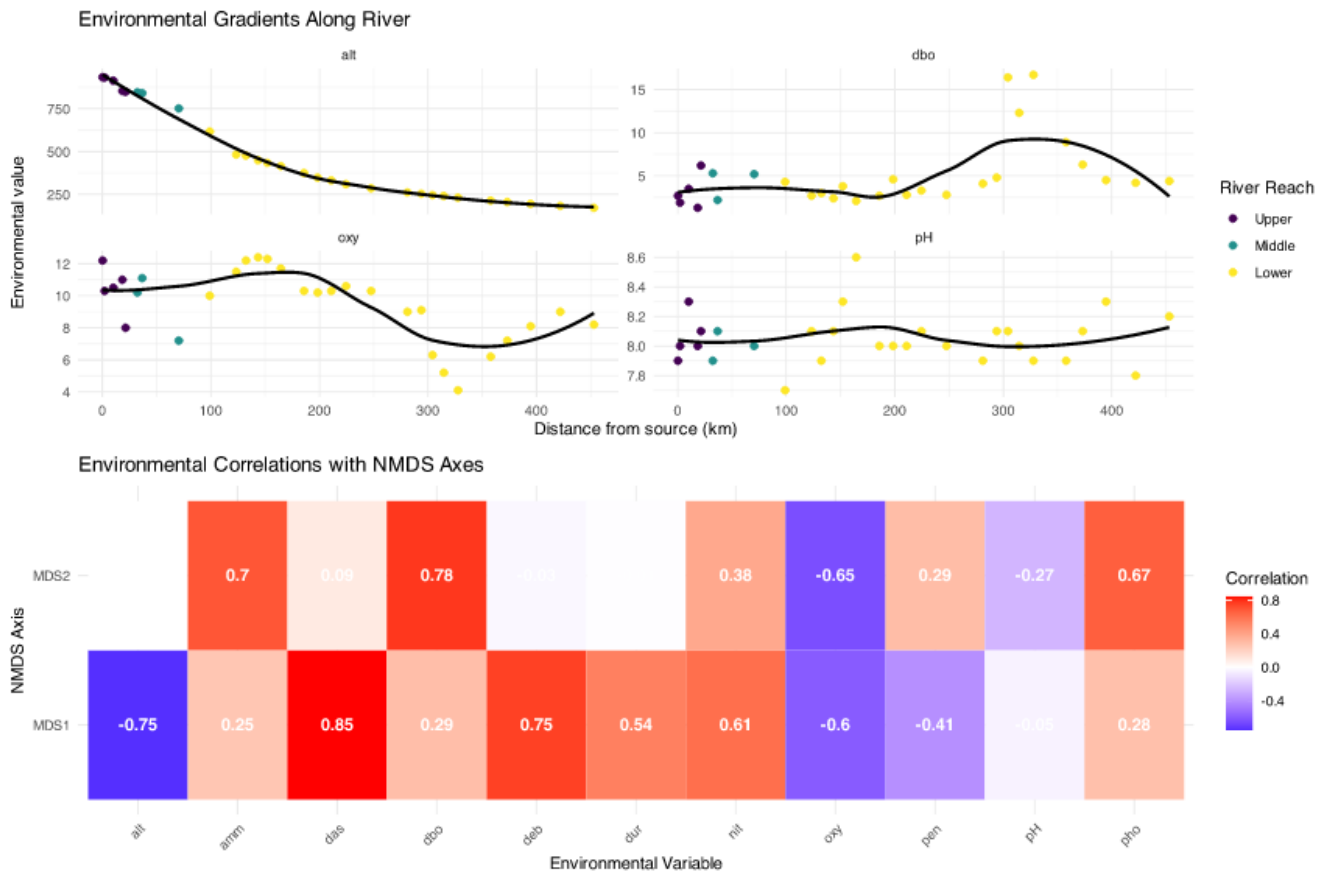
p2 <- nmbs_env_cor %>%
  as.data.frame() %>%
  rownames_to_column("NMDS_axis") %>%
```

```

pivot_longer(cols = -NMDS_axis, names_to = "env_var", values_to = "correlation") %>%
ggplot(aes(env_var, NMDS_axis, fill = correlation)) +
geom_tile() +
geom_text(aes(label = round(correlation, 2)), color = "white", fontface = "bold") +
scale_fill_gradient2(low = "blue", mid = "white", high = "red",
                     midpoint = 0, name = "Correlation") +
labs(title = "Environmental Correlations with NMDS Axes",
     x = "Environmental Variable", y = "NMDS Axis") +
theme_minimal() +
theme(axis.text.x = element_text(angle = 45, hjust = 1))

```

p1 / p2



Key Environmental Patterns:

- **Distance from source (das):** Strong correlate with community change
- **Oxygen (oxy):** Decreases downstream, affects fish communities
- **Biological oxygen demand (dbp):** Increases downstream (pollution indicator)
- **Altitude (alt):** Decreases downstream, associated with temperature changes

Summary and Conclusions

Key Findings from Today's Analysis

NMDS Results:

- **Stress = 0.074:** Excellent representation
- **Clear gradient** from upper to lower river reaches
- **Within-reach similarity** > between-reach similarity

PERMANOVA Results:

- **Highly significant** differences between reaches ($p < 0.001$)
- **River reach explains substantial variance** in community composition
- **All pairwise comparisons significant** after correction

ANOSIM Results:

- **R = 0.119**: Excellent separation
- **Confirms PERMANOVA findings** with different approach
- **Strong within-group coherence**

Environmental Drivers:

- **Distance from source**: Primary gradient
- **Dissolved oxygen**: Decreases downstream
- **Biological oxygen demand**: Increases downstream
- **Multiple correlated factors** drive community change

Biological Interpretation:

- **River continuum concept supported**
- **Progressive community change** from source to mouth
- **Environmental filtering** shapes community assembly
- **Pollution gradient** evident in lower reaches

Take-Home Messages

NMDS (Non-metric Multidimensional Scaling):

- Visualizes community dissimilarity in 2D/3D
- Preserves rank order of distances (non-metric)
- No distributional assumptions
- Stress < 0.2 for acceptable solutions
- Great for exploring ecological gradients

PERMANOVA (Permutational MANOVA):

- Tests for differences in group centroids
- Uses distance matrices, not raw data
- No distributional assumptions
- Can handle complex designs
- Check dispersion homogeneity assumption

ANOSIM (Analysis of Similarities):

- Tests group separation using rank distances
- R-statistic: -1 (no separation) to $+1$ (complete)
- More robust to outliers than PERMANOVA
- Simpler but less powerful
- Good for exploratory analysis

Environmental Drivers:

- Use `envfit()` to correlate environment with ordination
- Identifies which variables explain community patterns
- Helps understand ecological mechanisms
- Essential for management implications

Best Practices Summary

Analysis Workflow:

1. Explore your data first

- Check for outliers and zeros
- Understand your sampling design
- Choose appropriate distance measure

2. Run NMDS for visualization

- Try multiple random starts
- Check stress values
- Interpret gradients carefully

3. Test hypotheses with PERMANOVA

- Check dispersion homogeneity first
- Include effect sizes in results
- Run pairwise tests if needed

Common Pitfalls to Avoid:

Don't:

- Ignore stress values > 0.2
- Forget to check PERMANOVA assumptions
- Report only p-values without effect sizes
- Over-interpret NMDS axes as meaningful
- Use these methods on inappropriate data

Do:

- Use multiple random starts for NMDS
- Check assumption violations
- Include biological interpretation
- Consider data transformations
- Validate results with different methods